

Exercise 4: Arrange-Act-Assert (AAA) Pattern, Test Fixtures, Setup and Teardown Methods in JUnit

BankAccount.java:

```
package com.example;

public class BankAccount {
    private double balance;

    public BankAccount(double initialBalance) {
        this.balance = initialBalance;
    }

    public void deposit(double amount) {
        balance += amount;
    }

    public double getBalance() {
        return balance;
    }
}
```

BankAccountTest.java:

```
package com.example;

import static org.junit.Assert.assertEquals;
import org.junit.Before;
import org.junit.After;
import org.junit.Test;

public class BankAccountTest {
    private BankAccount account;

    @Before
    public void setUp() {
        // This is part of 'Arrange' but shared across tests
        account = new BankAccount(100.0);
        System.out.println("Before Test: Account initialized with $100.");
    }

    @Test
    public void testDeposit() {
```

```

// Arrange
double depositAmount = 50.0;

// Act
account.deposit(depositAmount);

// Assert
assertEquals(150.0, account.getBalance(), 0.001);
System.out.println("Test: Deposit logic verified.");
}

@After
public void tearDown() {
    account = null;
    System.out.println("After Test: Cleaning up account object.");
}
}

```

Output:

```

● adityahargunani@Adityas-MacBook demo % mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:demo >-----
[INFO] Building demo 1.4
[INFO] from pom.xml
[INFO]
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.0.2:resources (default-resources) @ demo ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory /Users/adityahargunani/Downloads/Exercise 4_ Arrange-Act-Assert (AAA) Pattern/demo/src/main/resources
[INFO]
[INFO] --- compiler:3.8.0:compile (default-compile) @ demo ---
[INFO] Changes detected - recompiling the module!
[INFO] Compiling 1 source file to /Users/adityahargunani/Downloads/Exercise 4_ Arrange-Act-Assert (AAA) Pattern/demo/target/classes
[INFO]
[INFO] --- resources:3.0.2:testResources (default-testResources) @ demo ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory /Users/adityahargunani/Downloads/Exercise 4_ Arrange-Act-Assert (AAA) Pattern/demo/src/test/resources
[INFO]
[INFO] --- compiler:3.8.0:testCompile (default-testCompile) @ demo ---
[INFO] Changes detected - recompiling the module!
[INFO] Compiling 1 source file to /Users/adityahargunani/Downloads/Exercise 4_ Arrange-Act-Assert (AAA) Pattern/demo/target/test-classes
[INFO]
[INFO] --- surefire:2.22.1:test (default-test) @ demo ---
[INFO]
[INFO] T E S T S
[INFO]
[INFO] Running com.example.BankAccountTest
Before Test: Account initialized with $100.
Test: Deposit logic verified.
After Test: Cleaning up account object.
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.019 s - in com.example.BankAccountTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 1.102 s
[INFO] Finished at: 2026-06-23T16:17:04+05:30
[INFO]
○ adityahargunani@Adityas-MacBook demo %

```

